WOCSA-Sat-Corps

Experimental Telecommand and Tracking Protocol

Zigbee-Encapsulated 8-bit TC/TR Specification

Document Type	Experimental Specification
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This document is inspired by CCSDS and ECSS standards.
It is NOT CCSDS nor ECSS compliant and shall not be used for operational or real spacecraft systems.

Contents

1 Scope **2**

2 Applicable and Reference Documents **2**

3 Terms, Definitions and Abbreviations **2**

4 Protocol Overview **2**

5 Protocol Layering **3**

6 Telecommand Packet Format **3**

6.1 TC Packet Bit Allocation 3

6.2 TC Packet Bit Description 3

6.3 Target Field Encoding (bits b–c) 3

6.4 Command Field Encoding (bits d–e) 4

6.5 Value Field Encoding (bits f–g) 4

7 Tracking Packet Format **4**

7.1 TR Packet Bit Allocation 5

7.2 TR Packet Bit Description (Compact) 5

7.3 Source Subsystem Encoding (bits b–c) 5

7.4 Command / Data Type Encoding (bits d–e) 5

7.5 Value / Status Encoding (bits f–g) 5

8 Error Control Bit (bit h) **6**

A Example Telecommand Packet **6**

B Example Tracking Packets **6**

1 Scope

This document specifies the **WOCSA-Sat-Corps** protocol.

WOCSA-Sat-Corps is an experimental Telecommand (TC) and Tracking (TR) protocol designed for educational workshops, satellite simulators, and laboratory environments.

The protocol is intentionally simple to allow participants to decode telecommands and tracking data manually or programmatically.

2 Applicable and Reference Documents

- CCSDS 232.0 – TC Space Data Link Protocol (reference only)
- CCSDS 401.0 – Radio Metric Tracking (reference only)
- ECSS-E-ST-70-41 – Packet Utilization Standard (reference only)

3 Terms, Definitions and Abbreviations

TC	Telecommand
TR	Tracking
CRC	Cyclic Redundancy Check
PN	Pseudo-Noise

4 Protocol Overview

WOCSA-Sat-Corps uses a compact 8-bit command word to represent telecommands and tracking identifiers.

The protocol distinguishes between:

- Commands sent from ground to satellite (TC)
- Measurements sent from satellite to ground (TR)

5 Protocol Layering

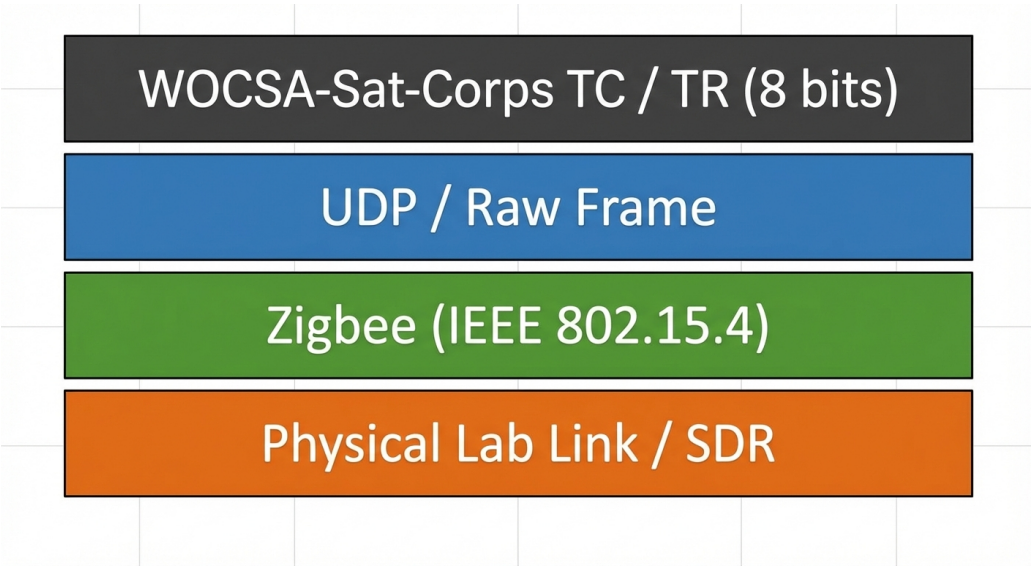

Figure 1: WOCSA-Sat-Corps protocol layering over Zigbee

6 Telecommand Packet Format

A WOCSA-Sat-Corps Telecommand (TC) packet shall consist of exactly one byte (8 bits).

Each bit has a predefined semantic meaning and shall be decoded according to the rules defined in this clause.

6.1 TC Packet Bit Allocation

The TC packet byte shall be structured as follows:

0	1 2	3 4	5 6	7
Dir	Target	Cmd	Value	EC

6.2 TC Packet Bit Description

Bit(s)	Name	Description
0	Direction (a)	0 = TC, 1 = TR
1–2	Target (b–c)	Subsystem addressed
3–4	Command (d–e)	Command identifier
5–6	Value (f–g)	Command parameter
7	Error Control (h)	XOR checksum bit

6.3 Target Field Encoding (bits b–c)

The Target field shall identify the subsystem addressed by the telecommand.

b c	Target Subsystem
00	System / General
01	Motor Control
10	GPS Subsystem
11	Reserved

6.4 Command Field Encoding (bits d–e)

The Command field shall define the action to be performed by the target subsystem.

Motor Control (Target = 01)

d e	Command
00	Stop motor
01	Increase power
10	Decrease power
11	Rotate motor

GPS Subsystem (Target = 10)

d e	Command
00	Access X coordinate
01	Access Y coordinate
10	Access Z coordinate
11	Reserved

6.5 Value Field Encoding (bits f–g)

The Value field shall provide a parameter associated with the command.

- For Motor Control commands, the Value field represents power level or rotation angle.
- For GPS commands, the Value field shall be ignored and set to zero.

f g	Motor Value
00	Low power / 15 degrees
01	Medium power / 30 degrees
10	High power / 45 degrees
11	Maximum / Custom

7 Tracking Packet Format

A WOCSA-Sat-Corps Tracking (TR) packet shall consist of exactly one byte (8 bits).

Each bit shall be interpreted according to the rules defined in this clause.

7.1 TR Packet Bit Allocation

The TR packet byte shall be structured as follows:

0	1 2	3 4	5 6	7
Dir	Src	Cmd	Val	EC

7.2 TR Packet Bit Description (Compact)

Field	Description
Dir (a)	Direction: 0 = TC, 1 = TR
Src (b–c)	Source Subsystem (which generated the TR)
Cmd (d–e)	Command / Data Type: GPS → X/Y/Z, Motor → ACK/ERROR
Val (f–g)	Value / Status: GPS → quantized coordinate (scaled), Motor → ACK/ERROR
EC (h)	Error Control: XOR of bits a through g

7.3 Source Subsystem Encoding (bits b–c)

b c	Subsystem
00	System / General
01	Motor Control
10	GPS Subsystem
11	Reserved

7.4 Command / Data Type Encoding (bits d–e)

GPS Subsystem (b c = 10)

d e	Requested Coordinate
00	X coordinate
01	Y coordinate
10	Z coordinate
11	Reserved

Motor Control (b c = 01)

d e	Motor Status Report
00	ACK (command executed successfully)
01	ERROR (command failed)
10	Reserved
11	Reserved

7.5 Value / Status Encoding (bits f–g)

GPS Subsystem (b c = 10) The Value field shall provide a quantized representation of the requested coordinate:

f g	Coordinate Value
00	Minimum scale (0 * 100 units)
01	Low (1 * 100 units)
10	Medium (2 * 100 units)
11	High (3 * 100 units)

Motor Control (b c = 01) The Value field shall indicate the outcome of the last command:

f g	Status
00	ACK
01	ERROR
10	Reserved
11	Reserved

8 Error Control Bit (bit h)

The Error Control bit shall be computed as the XOR of bits a through g:

$$h = a \oplus b \oplus c \oplus d \oplus e \oplus f \oplus g$$

Packets with an invalid Error Control bit shall be discarded.

A Example Telecommand Packet

Example TC byte:

```
0b0 01 11 10 1
```

- a = 0 → Telecommand
- b c = 01 → Motor Control
- d e = 11 → Rotate motor
- f g = 10 → 45 degrees
- h = XOR checksum

B Example Tracking Packets

Motor TR Example 0b1 01 00 00 1

- a = 1 → TR
- b c = 01 → Motor Control

- d e = 00 → ACK
- f g = 00 → Status OK
- h = XOR checksum

GPS TR Example 0b1 10 01 10 1

- a = 1 → TR
- b c = 10 → GPS
- d e = 01 → Y coordinate
- f g = 10 → Medium value (scaled by 100)
- h = XOR checksum